

AMRITA KAUR BEDI

amritabedi27@gmail.com | (925) 905-6152 | in/amritakaurbedi | U.S. Permanent Resident

EDUCATION

University of California, Irvine

Expected June 2028

Bachelor of Science in Biomedical Engineering | Minor in Bioinformatics | Minor in Innovation & Entrepreneurship

Cumulative GPA: 3.957 | Dean's Honor List – 6 Quarters

TECHNICAL EXPERIENCE & PROJECTS

Applied Medical

June 2026 – Present

Quality Engineering Intern

- Perform risk assessments and develop technical justifications for initiating Corrective and Preventive Actions (CAPAs) in accordance with **ISO 13485** and **21 CFR Part 820** requirements
- Lead two CAPA investigations through document review, root cause analysis, and cross-functional process evaluation, presenting findings and corrective-action recommendations to senior management
- Redesign a package peel-test fixture in SolidWorks, create engineering drawings, and fabricate and validate the design to improve the consistency and reliability of FFS package seal-strength testing

Laparoscopic Trainer Lab Module Development

January 2026 – June 2026

Team Lead | UCI–Applied Medical Collaboration

- Designed and executed benchtop testing protocols to quantify laparoscopic tool-tissue interaction forces using Arduino-based data acquisition systems
- Applied **ISO 7741** and **FDA 510(k)** design-control principles to evaluate medical device performance and safety margins
- Co-authored a published [ASEE conference paper](#) and developed SOPs, instructional materials, and quantitative analyses for an ABET-accredited biomedical engineering laboratory course

PetalGrip: Redesigned Cervical Cancer Biopsy Forceps

April 2026 – June 2026

Design Engineer

- Developed a novel cervical cancer biopsy forceps concept in SolidWorks, creating detailed part models, assemblies, and engineering drawings for rapid prototyping
- Conducted iterative design optimization through SolidWorks simulations, prototype testing, and performance evaluation to improve functionality, manufacturability, and ergonomics
- Authored a comprehensive [engineering design report](#) detailing design rationale, prototyping methodology, testing results, material selection, manufacturing considerations, and cost analysis

Novel Suture Project

May 2025 – Present

Fabrication Lead

- Built and calibrated a custom PCB-based pressure sensor (range **0–183 kPa**, $R^2 = 0.997$, **0.046%** residual error) for biomedical force monitoring applications
- Conducted hardware validation using MATLAB, achieving **0.20 kPa** resolution and **26.5 dB** SNR, while evaluating repeatability, accuracy, and process performance
- Authored and implemented SOPs for fabrication, testing, and design verification under quality control standards
- Presented at the 2025 National BMES Conference and UCI Undergraduate Research Opportunities Program Symposium

Downing Lab at UCI

April 2025 – Present

Data Analytics Lead

- Investigated cytoskeletal changes in lung cancer cell lines following CSRP1 knockdown using immunofluorescence staining and quantitative data analysis; measured **20+** cell morphology parameters using ImageJ, Python, and R
- Developed a Python-based distance analysis pipeline to quantify spatial relationships between cellular structures from immunofluorescence imaging data
- [Presented](#) findings demonstrating statistically significant disruption ($p < 0.0001$) in actin organization following CSRP1 knockdown at the 2025 National BMES Conference

LEADERSHIP & INVOLVEMENT

Biomedical Engineering Society at UCI

January 2025 – Present

Project Coordinator

- Founded and manage the MedTech Founders Program, mentoring **20+** students from concept to prototype; **three** teams advanced to the semifinals of UCI entrepreneurship competitions
- Organized **3+** SolidWorks, Arduino, and programming workshops, strengthening design-for-manufacturability and rapid-prototyping skills for **50+** undergraduates
- Coordinated cross-functional events engaging **100+** students in hands-on engineering projects
- Designed a mobile app-controlled temperature probe integrating an ESP32, custom electronics, and a SolidWorks-modelled enclosure

SKILLS

Design & Prototyping: SolidWorks (CSWA Certified), Engineering Drawings, 3D Printing, PCB Design, Laser Cutting, PDMS

Programming & Data Analysis: Python, MATLAB, R, Arduino, ImageJ, SQL, Statistical Analysis, Data Visualization

Testing, Quality & Product Development: Sensor Calibration, Design Validation, CAPA, Root Cause Analysis